



Mobile Software Engineer Take-home

Hello from the Blackboard team!

This exercise will help us determine how you think through problems on real-world scenarios. We're not looking for the "right" solution, instead we're going to evaluate how you approach problems in code, user experience, design, architecture and workflows.

This take-home tries to emulate a real-world scenario inside Blackboard Studio.

Introduction

Blackboard Studio is working on its newest app: MealPrep; MealPrep is an app that helps people create tailored weekly meal plans based on their preferences in minutes.

Resources

Figma

[MealPrep Design.fig](#)

Font

[promo.zip](#)

Products Database

[product_catalog_en.json](#)

API Keys (for LLM workflows)

```
OPENAI-API-KEY: "sk-svcacct-Q1arx0YkJ2ofMAgIoYctwF4zmPwwi0QeDPDeeXuil0zW_bkmRobH-BKcCvxCl  
kYniMQ6akn3HPT3BlbkFJzA3e4SB1QlIWep6VEJfmI5lYsekpCWDv7PgRad00q1bAZ8y3GajrQZWqhhiBiFeCMtAM  
8yT7cA"
```

Scenario

Blackboard wants to create a POC of MealPrep to test the idea with an initial user-base. To do that, we only need the single core flow of the app.

The core flow of the app consists of 5 screens:

1. App lander
2. Budget selection
3. Dietary needs selection
4. Nutritional goals selection
5. Complete meal plan

The engineering team scraped all the Esselunga products from the OpenFoodFacts database and enriched them with prices, you'll find the complete database in the resources above. The items in the database have all the necessary properties to complete the development.

The design team started working on the design, so most of the screens are ready to be implemented pixel-perfectly. You'll find the design file in the resources above.

But the lead designer had some urgent priorities in a different team, and left the design file incomplete. You'll have to improvise small parts of the missing design

as you go.

Goal



"As a user, I want a quick and easy way to come up with a meal plan that fits my budget and preferences."

Your goal is to develop the flow as a standalone working app at the best of your abilities.

You can freely choose to develop the app in either React Native or SwiftUI, this won't matter in the evaluation.

Unknowns

The exercise contains many unknowns, including missing UI/UX specs.

Suggestion: when facing a decision always ask yourself: "how would I do it if it was my own app, and I was about to publish it?".

Use of AI tools

Use of AI tools is not only allowed, but encouraged.

You will be required to deliver a high-level overview of the tools you've used (agents, skills, MCPs, harnesses, etc.) as well as all your conversation logs to demonstrate your abilities in using the AI tools.

Screens details

01 Lander

This is the starting view you land on when opening the app.

Everything is made to be replicated pixel-perfectly apart from the assets (image + emojis) in the middle, where you have full ownership of switching the assets and/or adding any animations.

02 Budget Selection

This view serves the purpose of making the user select their desired weekly budget for their meal prep.

The final meal plan should reflect the selected budget.

The design is missing the main slider input required to edit the budget, it's completely up to you to implement it. The slider should give values in between [25, 150]. You are free to choose the slider steps.

03 Dietary Needs Selection

This view gives the user the opportunity to select any dietary needs they have. Some users could be vegetarians, others could be lactose intolerant.

This view has to be implemented pixel-perfectly.

It's up to you to decide the values and filters needed to retrieve the matching ingredients from the database.

The final meal plan should reflect the selected dietary needs.

04 Nutritional Goals Selection

This view gives the user the opportunity to select any nutritional preference.

This view has to be implemented pixel-perfectly.

It's up to you to decide the values and filters needed to retrieve the matching ingredients from the database.

The final meal plan should reflect the selected nutritional preference.

05 Weekly meal plan

Finally the user arrives at the last screen, where their whole meal plan lives.

The plan should be generated via an LLM workflow, API keys are provided above.



You'll have to come up with an LLM workflow that uses the configurations and ingredients the user selected as inputs and gives back the meal plan and recipes as an output.

You'll then have to format and display the recipes.

For the sake of simplicity, everything can be done client-side.

The user should be able to navigate through their meal plan easily, day by day.

Here you have the most freedom to develop the daily cards in any way you prefer, anything inside the big white card is completely up to you.

The only requirements are:

- The meal plan has to be 7 days.
- At least 1 meal per day.
- Include the meal name, preparation time, servings and price in any way you prefer.
- Include the necessary ingredients and the step by step recipe for each meal in any way you prefer.

Freedom of choices

Even though parts of the design are made to be replicated pixel-perfectly, never shy away from changing or improving anything from the design file.

Important: if you do end up diverging from the original design, you'll be required to submit your reasoning behind your changes.

Time constraints

You'll receive a deadline for the delivery of this assignment separately. There are no strict time constraints on the amount of time you have to dedicate to complete it.

Feel free to take time to polish, add details and come up with a result you're actually satisfied with.

Delivery

We expect the following assets at delivery:

1. The complete GitHub repository containing the code.
2. A screen recording of you going through the app (no face/voice needed).

3. Exported AI tools conversation logs.
4. A brief document containing: [A] Explaining any decision you took along the way that you want your evaluator to know (e.g. design decisions, workflows decisions, etc) [B] If you had to add one feature to this small flow/app, what would it be? And why?

Evaluation criteria

We'll evaluate the final delivery in multiple areas, in no particular order:

- Overall app development skills, focusing on code quality, frameworks and techniques used.
- Personal taste in app feel, animations and details.
- Ability to recreate pixel-perfect designs, from Figma to production.
- Ability to work around missing designs and unknown variables.
- Ability to use AI tools (Codex, Claude Code, etc).
- Implementation of LLM workflows.
- Problem solving in ambiguous scenarios.

Be prepared to show us your approach, workflows and answer any questions your evaluator might have in your technical interview.